

VMware ESXi

PHASE 1 PROFESSIONAL STUDY GUIDE

Virtualization • Hypervisors • ESXi • Management • vSphere Standard Networking

Built for: Fast revision, lab preparation, and interview practice

Includes: Visual poster, structured concepts, 20 Q&A;, networking flows, and memory sheet

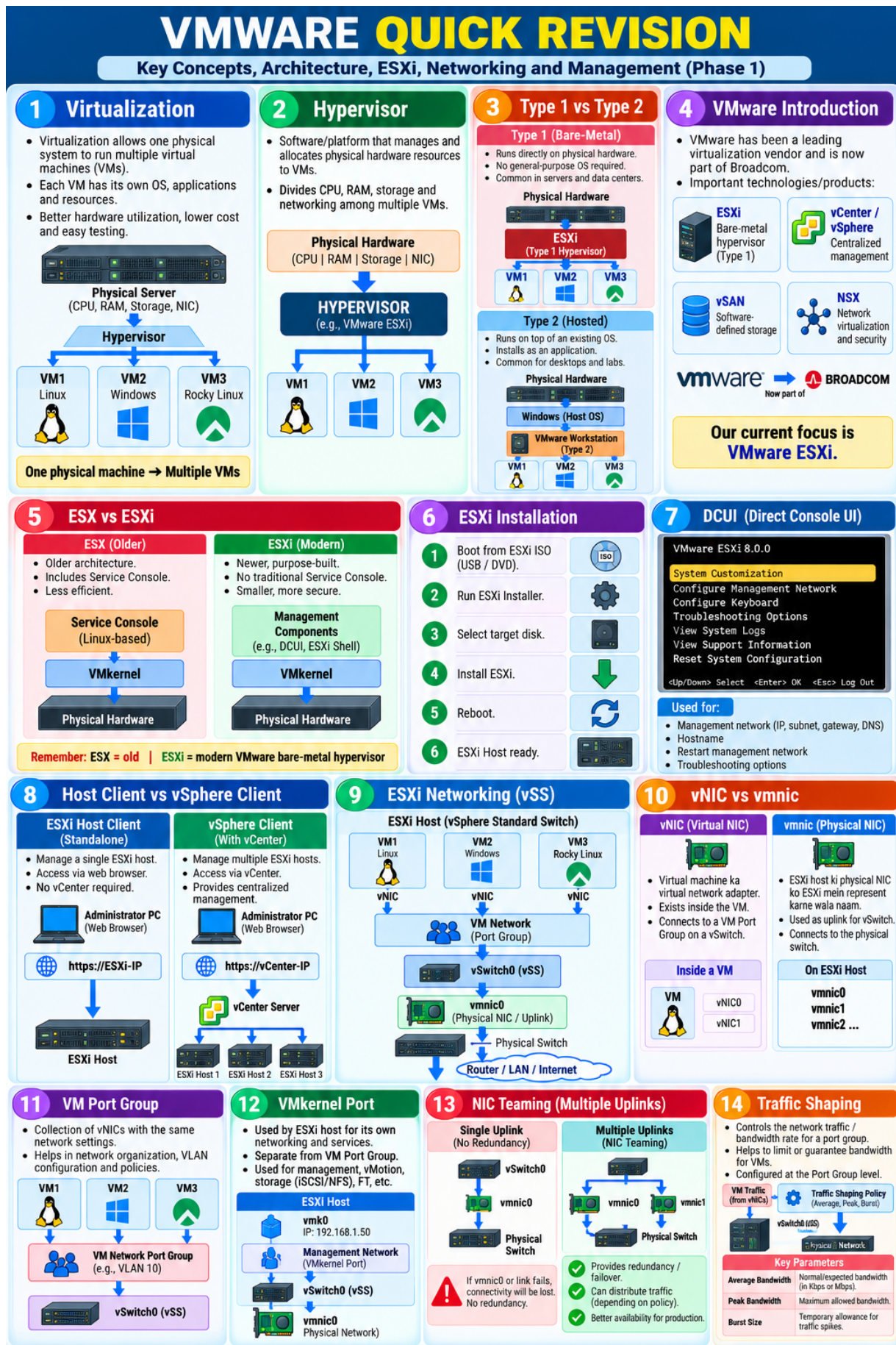
CORE TRAFFIC FLOW



Phase 1

1. Quick Revision Poster

Use this page for a 2-3 minute visual refresh before studying the detailed sections.



2. Study Roadmap

SECTION	WHAT YOU SHOULD KNOW
A. Virtualization Foundations	Virtualization, VM, Host vs Guest, Hypervisor, Type 1 vs Type 2
B. ESXi Architecture & Management	ESX vs ESXi, DCUI, VMkernel, browser-based host management
C. ESXi Networking	vSS, vNIC, vmnic, VM Port Group, VMkernel Port, NIC Teaming, Traffic Shaping
D. Interview Practice	20 concise questions and answers based on the Phase 1 material
E. Final Memory Sheet	One-page quick recall and the VM-to-physical-network path

How to study

1	Poster	Review the visual map.
2	Concepts	Understand the terms and architecture.
3	Traffic flow	Say the networking path aloud from memory.
4	Interview Q&A	Answer first, then check the model answer.

A. Virtualization Foundations

Build the mental model before learning ESXi.

Virtualization

One physical system can provide CPU, RAM, storage, and networking resources to multiple virtual machines.

Memory: Physical Server → Hypervisor → VM1 / VM2 / VM3

Virtual Machine (VM)

A software-based computer with virtual CPU, RAM, disk, NIC, operating system, and applications.

Memory: VM = vCPU + RAM + Virtual Disk + vNIC + OS

Host vs Guest

The host provides virtualization resources. The guest is the operating system running inside a VM.

Memory: Windows Host → VMware Workstation → Rocky Linux Guest

Hypervisor

The software/platform that manages physical hardware resources and makes them available to VMs.

Memory: CPU / RAM / Disk / NIC → Hypervisor → VMs

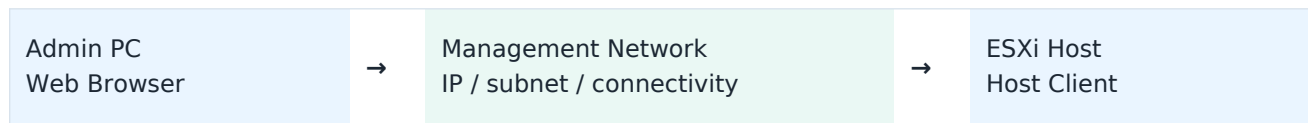
Type 1 vs Type 2 Hypervisors

	TYPE 1 - BARE-METAL	TYPE 2 - HOSTED
Placement	Runs directly on physical hardware	Runs on top of a host operating system
Example	VMware ESXi	VMware Workstation
Common use	Servers / data centers	Desktop / lab environments
Memory path	Hardware → ESXi → VMs	Hardware → Windows/Linux → Workstation → VMs

B. ESXi Architecture & Management

Know what ESXi is and how an administrator reaches it.

TOPIC	KEY POINT
ESX vs ESXi	ESX was the older architecture with a traditional Service Console. ESXi uses a newer purpose-built architecture.
DCUI	Direct Console User Interface - local/basic ESXi configuration and troubleshooting.
VMkernel	Core ESXi component handling CPU scheduling, memory, storage, networking, and VM access to hardware.
Host Client	A standalone ESXi host can be managed through a web browser using its management network.

Management connectivity

C. ESXi Networking

This is the most important Phase 1 networking section.

The complete VM-to-physical-network path

1	VM	Workload / guest system
2	vNIC	Virtual network adapter inside the VM
3	VM Port Group	VM connectivity and common network policies
4	vSwitch	Software-based Layer 2 virtual switch
5	vmnic	Physical NIC/uplink represented in ESXi
6	Physical Switch	External physical network switch
7	Physical Network	LAN / routed network as applicable

Golden distinction

VM needs network connectivity
Use a VM Port Group

ESXi host/service needs network connectivity
Use a VMkernel Port

TERM	PURPOSE
vSphere Standard Switch (vSS)	Software-based Layer 2 virtual switch inside an ESXi host.
vNIC	Virtual Network Interface Card assigned to a VM.
vmnic	Physical network adapter/uplink available to the ESXi host.
VM Port Group	VM connectivity plus common network configuration/policies.
VMkernel Port	ESXi host/service traffic such as management, vMotion, and certain storage traffic depending on configuration.
NIC Teaming	Multiple physical uplinks for redundancy/failover and policy-based traffic distribution.
Traffic Shaping	Controls network traffic rate/bandwidth using configured policy values.

D. Interview & Self-Test Practice

Try to answer each question aloud before reading the answer. The answers below stay close to the supplied Phase 1 study material.

Q1 What is virtualization?

Virtualization is a technology that allows us to use the CPU, RAM, storage, and networking resources of one physical server to run multiple Virtual Machines (VMs). Interview line: > One physical machine can provide resources to multiple independent virtual machines.

Q2 What is a VM?

A Virtual Machine (VM) is a software-based computer. It can have its own virtual CPU, RAM, disk, network adapter, operating system, and applications, while the underlying resources come from the physical host.

Q3 What is the difference between a Host and a Guest?

A Host is the system that provides resources for virtualization. A Guest is an operating system running inside a virtual machine.

Q4 What does a hypervisor do?

A hypervisor manages the physical server's CPU, memory, storage, and networking resources and makes those resources available to virtual machines.

Q5 What is the difference between Type 1 and Type 2 hypervisors?

A Type 1 hypervisor runs directly on physical hardware. A Type 2 hypervisor runs on top of an existing host operating system. Type 1 Type 2 Runs directly on hardware Runs on a host OS ESXi is an example VMware Workstation is an example Common in server/data-center Common in desktop/lab environments environments

Q6 Is ESXi Type 1 or Type 2?

VMware ESXi is a Type 1 (Bare-Metal) Hypervisor. It is designed to run directly on physical server hardware.

Q7 Is VMware Workstation Type 1 or Type 2?

VMware Workstation is a Type 2 hypervisor because it runs on top of a host operating system such as Windows or Linux.

Q8 What is the basic difference between ESX and ESXi?

ESX was VMware's older hypervisor architecture and included a traditional Linux-based Service Console. ESXi uses a newer, purpose-built architecture and does not include the traditional ESX Service Console. ESX = Older architecture ESXi = Modern VMware bare-metal hypervisor family

Q9 What is DCUI?

DCUI = Direct Console User Interface. It is the local console interface of an ESXi host and can be used for basic configuration and troubleshooting. Common tasks include: DCUI → Direct/local ESXi configuration

Q10 What is VMkernel?

VMkernel is a core component of ESXi. It handles important functions such as CPU scheduling, memory management, storage, networking, and access to hardware resources for virtual machines. > VMkernel = Core of ESXi.

Q11 What is a vSphere Standard Switch?

A vSphere Standard Switch (vSS) is a software-based Layer 2 virtual switch inside an ESXi host. It provides network connectivity for virtual machines and VMkernel networking and can connect to the physical network through physical uplinks.

Q12 What is a vNIC?

A vNIC is a Virtual Network Interface Card assigned to a virtual machine. Physical computer → NIC Virtual Machine → vNIC

Q13 What does vmnic0 represent?

vmnic0 represents a physical network adapter/uplink available to the ESXi host. Other examples include:

Q14 What is the purpose of a VM Port Group?

A VM Port Group provides network connectivity and common network configuration/policies for virtual machines. Port Groups can also be used for settings such as VLAN configuration and networking policies.

Q15 What is the purpose of a VMkernel Port?

A VMkernel Port/adaptor is used for the ESXi host's own network traffic and services. Examples can include: configuration Common interface names include: Example:

Q16 What is the difference between a VM Port Group and a VMkernel Port?

VM Port Group VMkernel Port
Used for VM traffic Used for ESXi host/service traffic
VM's vNIC connects to it Uses a VMkernel adapter (vmk)
Example: VM Network Example: Management Network
Guest workload networking Management, vMotion, storage-related host traffic, etc.

Q17 Why is NIC Teaming used?

NIC Teaming uses multiple physical NIC uplinks to provide redundancy/failover and can also distribute traffic according to the configured teaming/load-balancing policy. If one uplink fails, another uplink may maintain connectivity depending on the configuration.

Q18 What does Traffic Shaping control?

Traffic Shaping controls the rate/bandwidth of network traffic according to a configured policy. Important terms include: > Traffic Shaping = Controlling the network traffic rate according to policy.

Q19 Why is management networking required to access an ESXi host through a browser?

The administrator's computer and the ESXi host communicate over the network. Therefore, the ESXi host needs a properly configured management IP address, subnet configuration, and network connectivity. A default gateway is required when routing to other networks is necessary. DNS is useful for hostname-based name resolution. Example:

Q20 What path can VM traffic take to reach the physical network?

A typical VM traffic path from the virtual environment to the physical network is:
Quick Interview Revision # Self-Test Checklist
Make sure you can explain these concepts without looking at the answers: Phase 1 Interview & Self-Test Q&A; Complete

E. Final Memory Sheet

Use this page immediately before a lab, class, or interview.

TERM	REMEMBER
ESXi	Type 1 bare-metal hypervisor
VMware Workstation	Type 2 hosted hypervisor
VMkernel	Core ESXi component
DCUI	Direct/local ESXi configuration
vSwitch	Virtual Layer 2 switch
vNIC	VM's virtual network adapter
vmnic	ESXi physical NIC/uplink representation
VM Port Group	VM traffic
VMkernel Port	ESXi host/service traffic
NIC Teaming	Multiple uplinks / redundancy
Traffic Shaping	Network rate/bandwidth control

Must-memorize traffic path

VM → vNIC → VM Port Group → vSwitch → vmnic → Physical Switch → Network

Self-test checklist

☐ Virtualization	☐ VM
☐ Host vs Guest	☐ Hypervisor
☐ Type 1 vs Type 2	☐ ESX vs ESXi
☐ DCUI	☐ VMkernel
☐ vSphere Standard Switch	☐ vNIC
☐ vmnic	☐ VM Port Group
☐ VMkernel Port	☐ NIC Teaming
☐ Traffic Shaping	☐ Complete VM-to-physical-network flow